



Telco Customer Churn Analysis: Identifying Revenue Risk & Retention Opportunities Through Customer Segmentation

Executive Summary

Customer churn was concentrated in a few predictable high-risk segments, with **early-tenure Month-to-Month customers** showing the highest churn and revenue risk.

The most vulnerable customers typically had flexible contracts, early tenure, high charges, limited support, and stronger competitor alternatives.

The largest opportunity is to reduce churn through **first-180-day retention programs, proactive support, and contract upgrade campaigns.**

If churn is reduced by **15–25% in the highest-risk segment (Early-tenure Month-to-Month customers)**, the business could protect **₹300K–₹900K+ future revenue** while improving CLTV realization.

Problem Statement

The telecom company was experiencing customer churn across multiple customer groups but lacked clarity on:

- Which customers were most likely to churn
- Which churn segments created highest revenue loss
- Whether churn was driven by pricing, support, commitment, or competition
- Which retention actions would create maximum ROI

The objective was to convert raw churn data into clear business decisions.

Dataset

The dataset included customer-level telecom records with:

- Customer demographics
- Contract Type
- Tenure in Months
- Monthly Charges
- Internet Type
- Add-on Services
- Premium Tech Support
- Payment Method
- Customer Status (Stayed / Churned / Joined)
- CLTV
- Total Revenue
- Churn Category
- Churn Reason

Tools Used

- **Excel** -Data Cleaning, Data Manipulation
 - **SQL** – Segmentation, KPI calculations, churn analysis
 - **Power BI**– Dashboarding & visualization
 - **Business Analytics** – Revenue risk modelling, customer retention strategy
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Methodology

- The project used a segmentation-based churn analysis approach.
- Insights were prioritized using both churn rate and segment size to avoid over-interpreting small customer groups.
- Small sample segments were treated as directional indicators rather than final conclusions.

Core Metrics Created:

- Churn Rate
- Churned Customers
- Churned CLTV
- Realized Revenue
- Potential Revenue Loss = $CLTV - \text{Revenue Earned}$
- Revenue Risk %

Segmentations Performed:

- Contract × Tenure
 - Contract × Tenure × Monthly Charges
 - Contract × Tenure × Premium Support
 - Contract × Tenure × Internet Type
 - Contract × Tenure × Payment Method
 - Churn Category
 - Churn Reason
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Key Questions

1. Which customers churn the most?
 2. Which churn segments create highest future revenue loss?
 3. Does customer commitment reduce churn?
 4. Is churn driven by price, service quality, or competitors?
 5. Which business actions should be prioritized first?
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Findings

Finding 1: Contract Type × Tenure

Early-tenure Month-to-Month customers (1–6 months) were the most critical churn segment.

- 81.33% churn rate
- ₹3.0M potential revenue loss
- 94.84% CLTV at risk

Long-term contracts showed much lower churn.

Finding 2: Monthly Charges

High charges alone did not create churn.

But when combined with:

- Month-to-Month contracts
 - Early tenure
- the segment reached:
- ~90% churn
 - ₹1.1M potential revenue loss
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Finding 3: Premium Tech Support

For new Month-to-Month customers:

- Without support = **82.96% churn**, creating ₹2.7M potential revenue loss with 95% CLTV at risk.
- With support = **67.65% churn**

Support significantly improved retention.

Finding 4: Internet Type

Fiber Optic customers were highly valuable but vulnerable only when new and on flexible contracts.

- ~91% churn
 - 1.6 million potential revenue loss
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Finding 5: Payment Method

Highest-risk segment:

- Month-to-Month
 - 1–6 months
 - Bank Withdrawal
 - 88.05% churn
 - ₹2.01M potential revenue loss
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Finding 6: Churn Reasons

Main stated churn reasons among Month-to-Month customers:

- Better competitor devices/offers = 568 churns
- Poor support attitude = 284 churns

This indicates churn was influenced by both:

- External competitor value gaps
- Internal service experience issues

As these reasons were self-reported, they should be treated directionally rather than as absolute truth.

Finding 7: Satisfaction Score Data Check

Satisfaction score distribution suggested possible data leakage or post-churn survey timing, so it was excluded as a predictive churn driver.

9. Actions/Recommendations

Priority 1: Save New Month-to-Month Customers

Launch 180-day retention program: By First bill education ,30/60/90-day check-ins, Immediate support, Usage guidance

Priority 2: Upgrade Contracts

Target retained (7-12 months) Month-to-Month customers with: Annual discounts, Price-lock plans, Loyalty rewards

Priority 3: Improve/Add Premium Support Experience: Improve Premium Support for early-tenure Month-to-Month customers through proactive onboarding, faster resolution, trained agents, and recovery callbacks.

Priority 4: Competitor Défense Strategy

Retain at-risk customers with targeted bundles, better-value plans, device upgrade offers and save offers before cancellation.

Regularly research competitor pricing, devices, and promotions, and redesign or remove ineffective internal offers.

Run win-back campaigns for churned customers using rejoin discounts, rewards, or one free month.

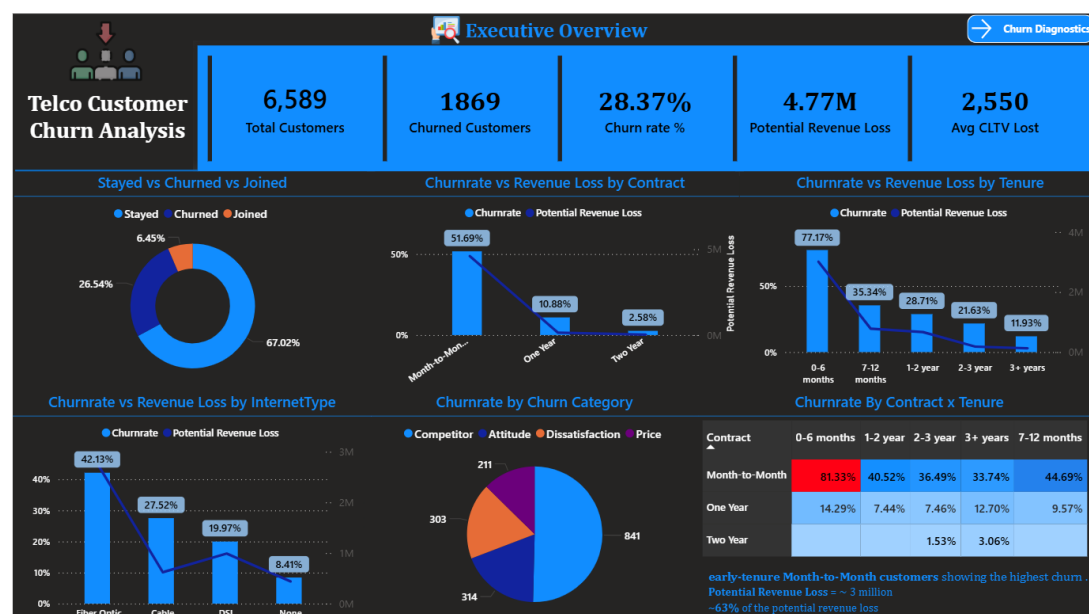
10. Business Impact

Successful execution could:

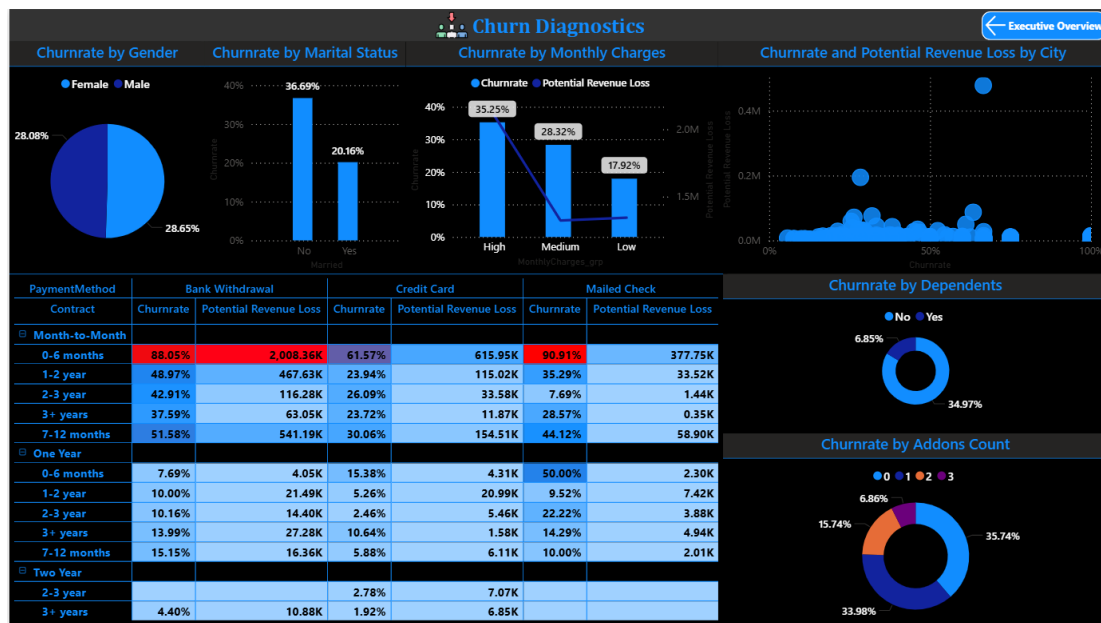
- Reduce churn in highest-risk segments
- Protect ₹300K–₹900K+ future revenue if churn is reduced by 15–25% in high-risk Month-to-Month early-tenure segment.
- Improve CLTV realization.
- Increase annual contract adoption
- Reduce acquisition waste
- Improve customer satisfaction and loyalty

11.Dashboard

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12. Conclusion

This project showed that churn was not random—it was concentrated in predictable customer groups.

Customers were most likely to churn when they had:

- Flexible Month-to-Month contracts
- Early tenure
- Weak onboarding
- High charges
- No support
- Better competitor alternatives

By combining churn analytics with revenue risk modelling, the company can shift from reactive churn reporting to proactive customer retention strategy.